

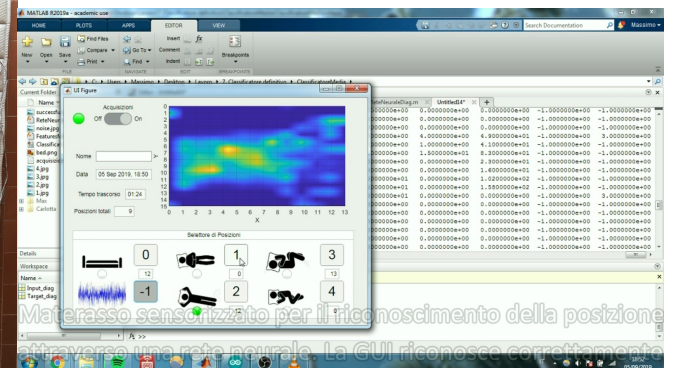
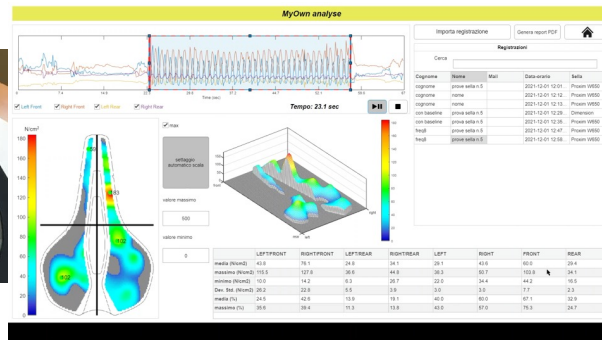
Bio-Inspired Resistive Sensors

Alessandro Tognetti

Research activity: Piezoresistive pressure sensors for Wearables

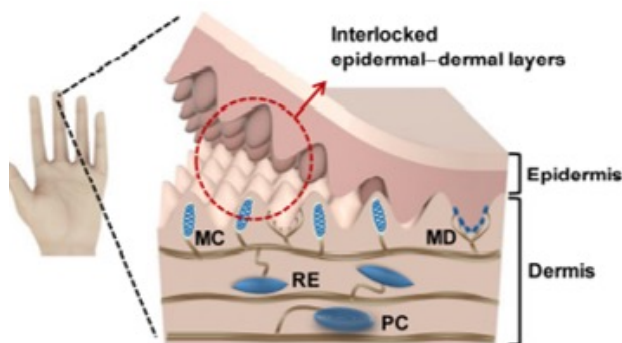


Dynamic analysis
(gait)



Materasso sensore per il riconoscimento della posizione
 attraverso una rete neurale. La GUI riconosce correttamente

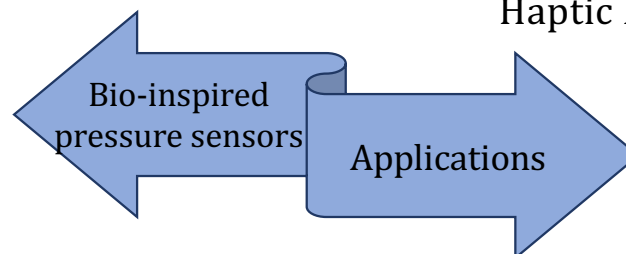
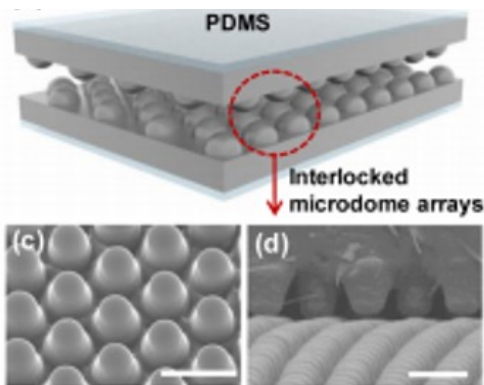
Research activity: Piezoresistive tactile sensors



Haptic Feedback Devices



Teleoperated Surgical Tools

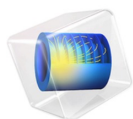


Wearable

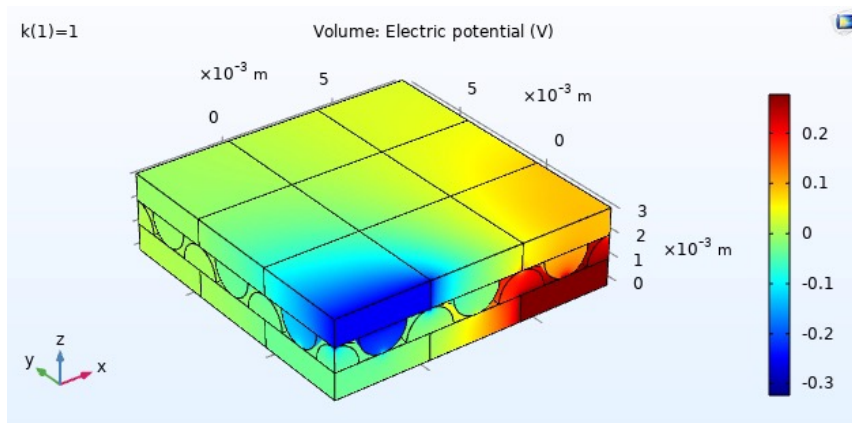
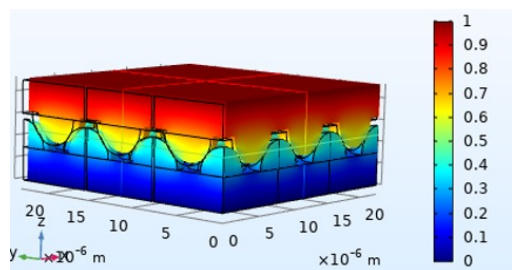


Prosthetics

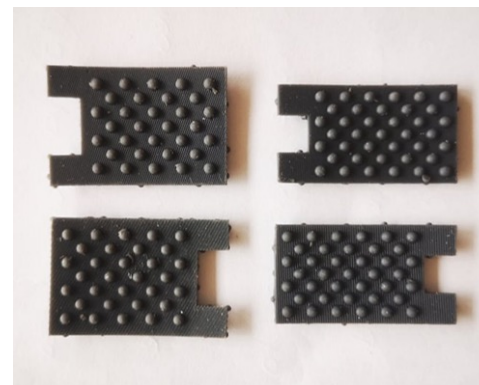
Research activity: Piezoresistive tactile sensors



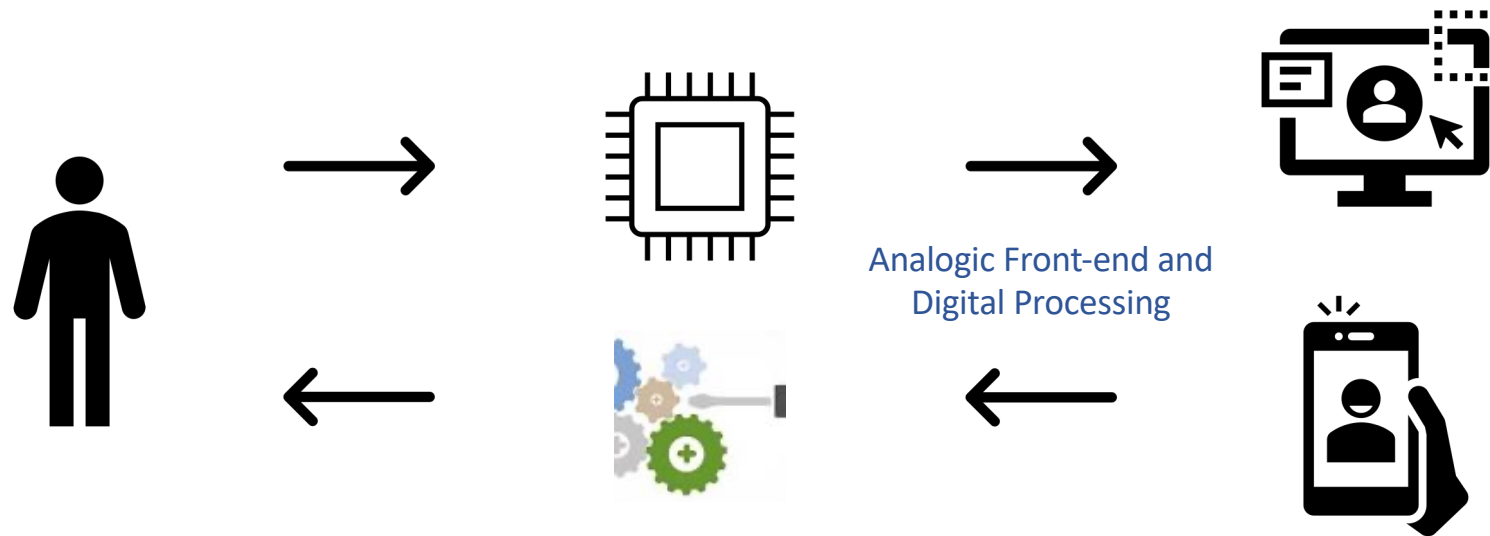
FEM simulations



Fabrication



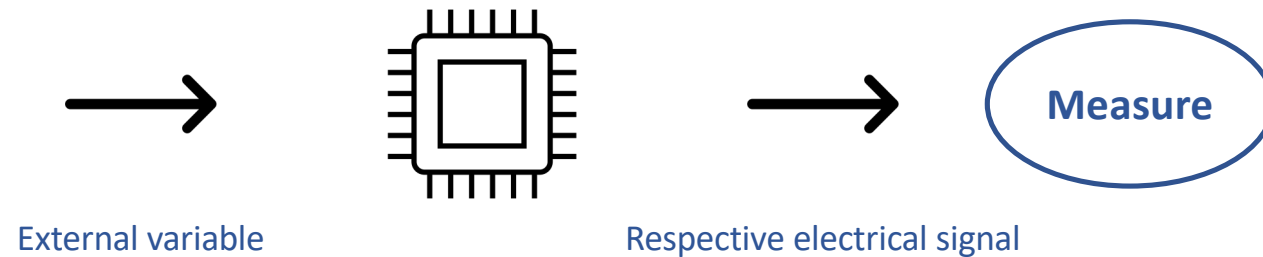
What is a sensor?



A sensor is a **transducer** that converts **physical variables** into a form that can be processed more easily and quickly, i.e., an **electrical signal**.

Goal: objective evaluation of the physical variable

What is a sensor?



Non-invasive → signal extracted without creating trauma, injury or alterations in vital parameters

Invasive

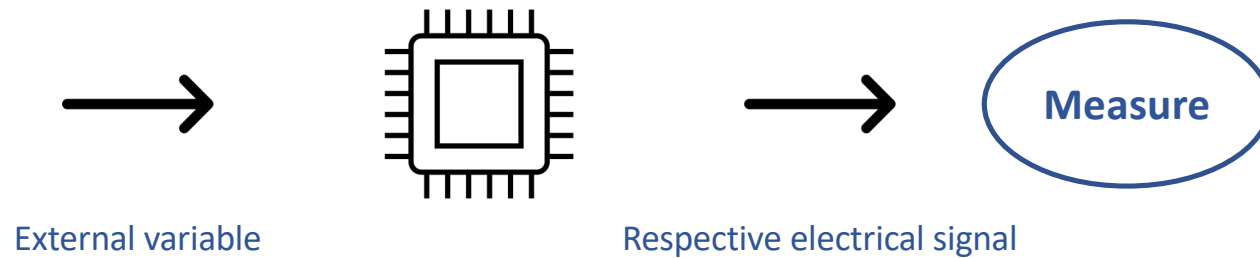
Active → in which the signal that will be processed later is emitted by the equipment itself

Passive → in which the origin of the signal to be elaborated derives spontaneously and directly from the organism

Direct → detection of physical and chemical quantities of interest by simply comparing them with reference values

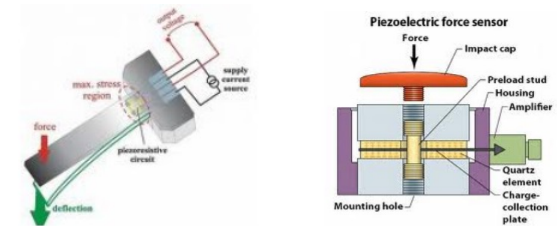
Indirect → based on mathematical models between the biomedical variable of interest and the quantity detected by the sensor

What is a pressure sensor?

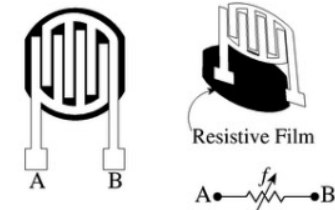


A sensing element that converts **applied force** into:

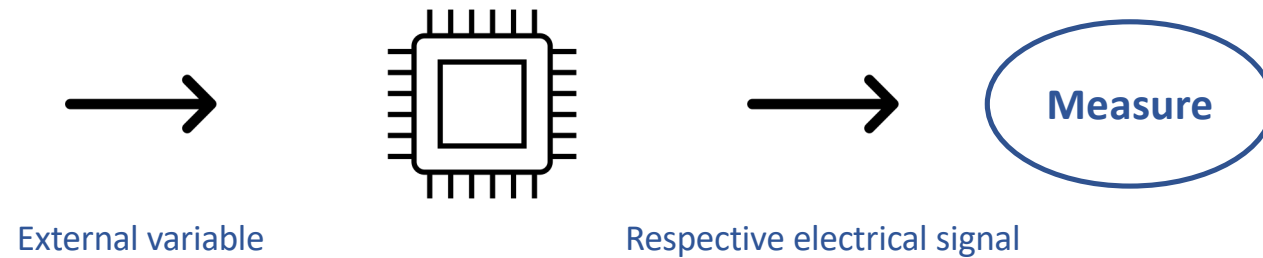
- mechanical displacement: es. Strain of an elastic element
- change in electrical magnitude: es. resistance or capacity



Given the surface area on which the force impresses, the **applied pressure** [Pa] (force per unite area [N/m^2]) can be derived.

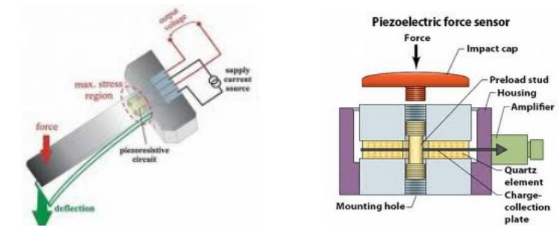


What is a pressure sensor?



Main applications

- Detect and measure the relative change in applied force or load
- Detect and measure the frequency of change in applied force
- Identify specific force values (thresholds) to trigger specific actions
- Detecting contact or touch of the sensor surface



Pressure sensors: The piezoresistive effect

Among the great variety of pressure sensing mechanisms (including piezoresistivity, piezoelectricity, capacitance and triboelectricity) **piezoresistive pressure sensors** have been widely used because of:

- Simple device structure and electronic interface
- Easy signal processing
- Good sensitivity
- Low-cost fabrication process

Pressure measurement consists essentially in a **change in resistance** generated by an applied mechanical stress, such as pressure or bending.

$$R = \frac{\rho l}{A}$$

Macroscopic effect (Metal conductors)

l → conductor length
 A → transverse section area
 ρ → material resistivity [$\Omega \cdot \text{cm}$]

Microscopic effect
(Semiconductors, es. silicon)

Pressure sensors: The piezoresistive effect

The piezoresistive effect is based on the shift in the atomic positions, which directly affects the electric charge transport and bandgaps, making it easier or harder for the electrons to be raised in the conduction band.

Piezoresistive materials need to be supplied by an external voltage source to read the change in resistance.

Characterizing the change in resistivity when a stress is applied, the **piezoresistive coefficient** can be derived.

Given:

Δ tensor for the relative change in resistivity/resistance

σ the stress tensor

The stresses on a solid can be expressed in form of a 2nd rank tensor with nine components (3 normal and 6 shear) which can be further simplified to six independent components (**3 normal and 3 shear**) using the conditions of force balance.

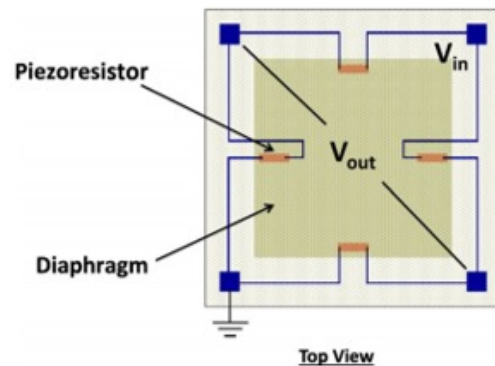
The relationship between these two can be expressed as:

$$\Delta = (\pi)(\sigma)$$

where π is the piezoresistive coefficient.

$$\begin{bmatrix} \Delta_1 \\ \Delta_2 \\ \Delta_3 \\ \Delta_4 \\ \Delta_5 \\ \Delta_6 \end{bmatrix} = \begin{bmatrix} \pi_{11} & \pi_{12} & \pi_{13} & \pi_{14} & \pi_{15} & \pi_{16} \\ \pi_{21} & \pi_{22} & \pi_{23} & \pi_{24} & \pi_{25} & \pi_{26} \\ \pi_{31} & \pi_{32} & \pi_{33} & \pi_{34} & \pi_{35} & \pi_{36} \\ \pi_{41} & \pi_{42} & \pi_{43} & \pi_{44} & \pi_{45} & \pi_{46} \\ \pi_{51} & \pi_{52} & \pi_{53} & \pi_{54} & \pi_{55} & \pi_{56} \\ \pi_{61} & \pi_{62} & \pi_{63} & \pi_{64} & \pi_{65} & \pi_{66} \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{bmatrix}$$

Pressure sensors: Classic piezoresistive



The basic form of a piezoresistive pressure sensor is based on:

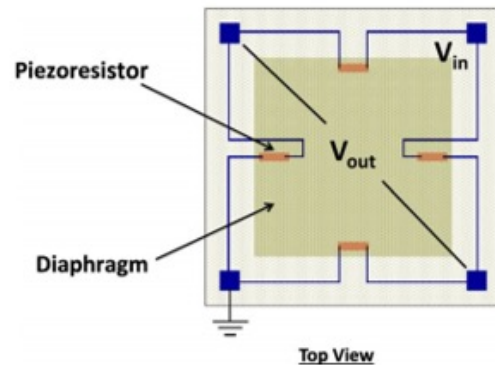
- a **diaphragm**, shaped in a rectangular or circular form and subjected to the input pressure load, which is transformed into the stress field
- **4 piezoresistors**, placed on the top of the diaphragm, arranged in a **Wheatstone bridge arrangement**, typically made of silicon

Initially, the resistances of all the resistors are equal. Then, the resistance of two of the piezoresistors increases and the other two decreases, making the bridge unbalanced and giving an output voltage.

The acquisition system works on the nodes: input current or voltage is applied on two nodes and the output voltage is measured at the other two.

The Wheatstone bridge arrangement helps in compensating the fluctuations of the resistors due to external factors in the measure, such as the temperature coefficient of resistance.

Pressure sensors: Classic piezoresistive



Very important to get control of the diaphragm dimensions while fabricating it, to find and realize the **optimal trade-off** between cost and requested dimensions.

Tunable parameters

Position of the piezoresistors on the diaphragm: best results in term of sensitivity and maximization of piezoresistive effect are obtained when the elements are placed in the high stress regions.

Diaphragm size and thickness:

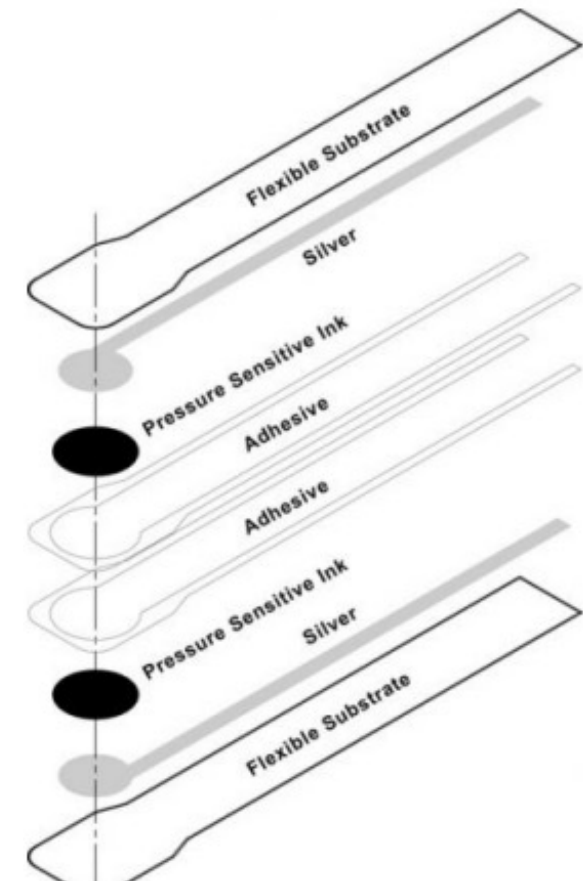
- Thinner diaphragms are more sensitive than thicker ones but more difficult to realize.
- Larger diaphragms help increasing the sensitivity but need also a bigger device.
- At thinner diaphragms, a bigger size leads to degradation in linearity.

Pressure sensors: Classic piezoresistive for flexible applications

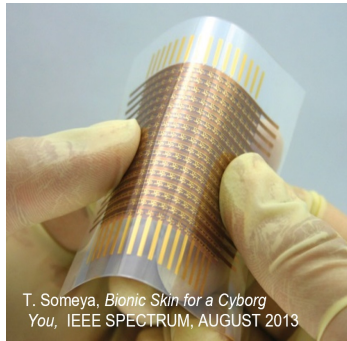
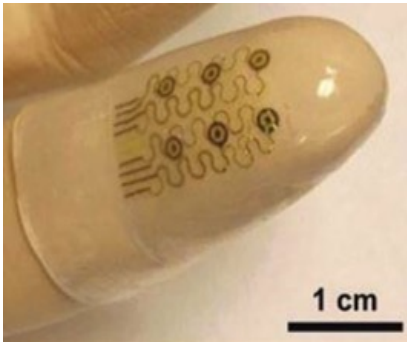


The FlexiForce sensor is constructed to generate variations of its nominal (rated) resistance due to the applied force

- Zero Force = no load applied → High resistance sensor (above $5\text{M}\Omega$)
- Force applied to the sensor → Sensor resistance decreases
- Measured force/resistance relationship is inversely proportional



Pressure sensors: Piezoresistive effect for flexible applications



First prototypes of flexible pressure sensors for tactile applications were realized with piezoresistive pressure sensors, **integrating rigid elements on elastic substrates**

- × low mechanical reliability
- × complex and expensive fabrication methods

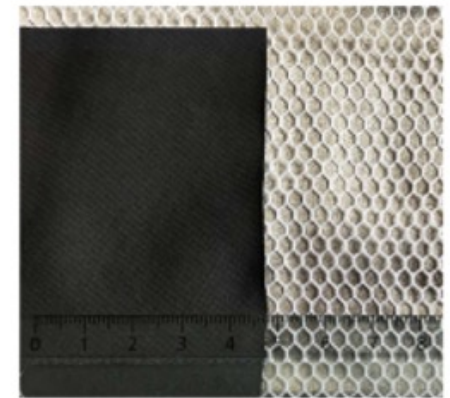
Piezoresistive fabric sensors

Combination of different materials that constitute multilayer flexible sensor:

- Layer/s made of a highly electrically conductive fabric
- Layer realized with a high-resistance piezoresistive fabric
- In the middle a spacer layer (e.g., a honeycomb mesh)

In touch only when an mechanical input is applied over the sensor.

Other benefits of the structure are its flexibility, the possibility of being cut to arbitrary shapes and being built with low-cost and available materials, with low effort.



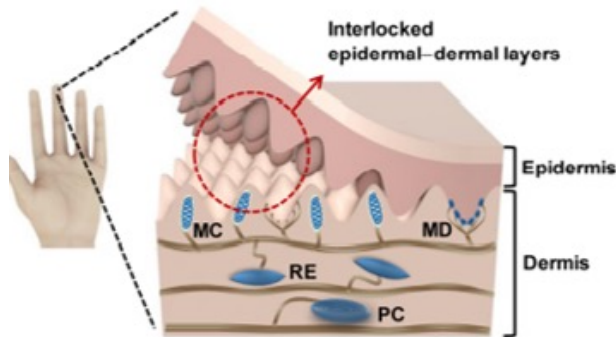
Bio-Inspired Piezoresistive Sensors



M.Sc. Bionics
Engineering



Bio-Inspired interlocked piezoresistive sensors



Intermediate ridges at the epidermal – dermal junction allowing **stress concentration** at the ridge tips where mechanoreceptors are located.

Transduction and **magnification** of the mechanical stimuli

Two layers **pressure sensor** with **conductive interlocked micro-domes**:

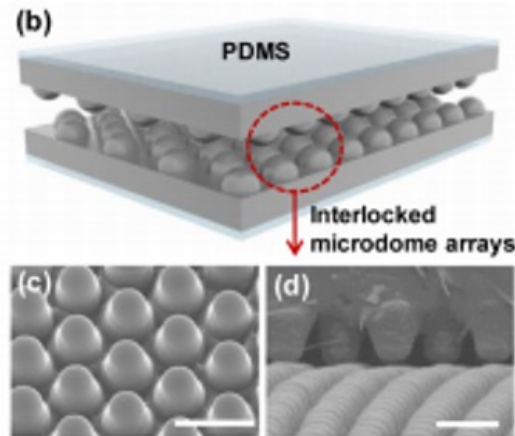
- **Flexibility** and **adaptability**: reducing the alteration of the host system stiffness
- **Global piezoresistive** effect: contact areas between the domes allow the onset of conductive paths where current flows, depending on the external pressure applied
- **Lightness** and **reduced encumbrance**

Bio-Inspired interlocked piezoresistive sensors

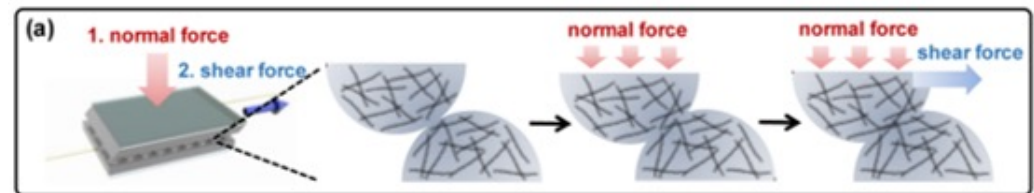
MICRO-STRUCTURES: Micro-Domes

Carbon nanotubes (CNT) composite conductive elastomeric films

Polymeric matrix: **PDMS** (Polydimethylsiloxane)
[3 μm height x 4 μm diameter]



PERFORMANCE: Stress concentrations



Detection of mechanical stimuli applied along 3 different principal directions

- Normal
- Stretch
- Bending / Shear Stress

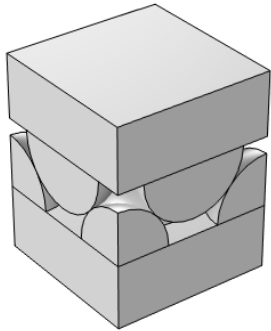
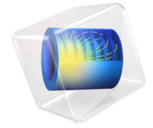
How sensors are normally characterized?

Tactile-Direction-Sensitive and Stretchable Electronic Skins Based on Human-Skin-Inspired Interlocked Microstructures

Jonghwa Park,^{1,2} Youngoh Lee,^{1,2} Jaehyung Hong,² Youngsu Lee,¹ Minjeong Ha,¹ Youngdo Jung,³ Hyuneui Lim,³ Sung Youb Kim,² and Hyunhyub Ko^{*,1}

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Bio-Inspired interlocked piezoresistive sensors



Objective: apply a displacement, derive the pressure and the current that flows in the device. Finally derive the sensor static characteristic: Electrical resistance vs. Pressure

SETTINGS & BOUNDARY CONDITIONS

Domains: Solid Mechanics, Electric Currents;

Study: Stationary

Elementary unit repeated in a 3D Array with *Periodic Condition*

PDMS with **CNT** dispersion

Young Modulus	2.6 [Mpa]
Conductivity	0.5 [S/m]

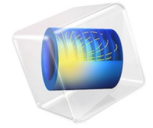
Superior layer: prescribed displacement and Electric Potential (1V)

Inferior layer: fixed constraint and electrical ground

Contact Pairs: *Penalty mode* and ideal contact impedance

Mesh refinement in contact surfaces

Bio-Inspired interlocked piezoresistive sensors



Model Definition

Model wizard

- Open the model wizard and select 3D dimension
- In the Select Physics panel, choose Structural mechanics|Solid mechanics. Click “Add”.
- In the Select Physics panel, choose AC/DC|Electric Currents. Click “Add”.
- Click the Study arrow to open the Select Study panel. Select Stationary and click “Done”.

This will exit the Model Wizard, displaying the main COMSOL interface.

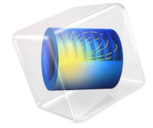
Parameters			
Label: Parameters 1			
Parameters			
Name	Expression	Value	Description
w_ly	7.5 [um]	7.5E-6 m	length of the layer
w_lx	7.5[um]	7.5E-6 m	width of the layer
sp	0.8 [um]	8E-7 m	space between layers
r_d	2.5 [um]	2.5E-6 m	radius of a microdome
h_l	3 [um]	3E-6 m	height of the layer
disp_max	(0.5)[um]	5E-7 m	maximum displacement
disp	0 [um]	0 m	initial displacement
steps	30	30	displacement steps

Global definitions

In the model builder select Parameters 1, under the Global definitions node. Specify the parameters as in the figure.

Dependent variables:
displacement and electric
potential

Geometry



bottom structure

Right-click Geometry 1 and select Block. Specify the parameters in Size and Shape as (w_lx, w_ly, h_l) .

Right-click Geometry 1 and select sphere. Specify r_d as radius and set the z-position to h_l

Right-click Block 1 and select Duplicate. Change the z-position to h_l

Right-click Geometry 1 and select Booleans and Partitions|Intersection. For the input objects, select the sphere (sph1) and the last block (blk2).

Repeat the previous 3 steps changing each time the position of the sphere center with the following coordinates: $(w_lx, 0, h_l)$; (w_lx, w_ly, h_l) ; $(0, w_ly, h_l)$; $(w_lx/2, w_ly/2, h_l)$.

Right-click Geometry 1 and select Booleans and Partitions|Union (name: bottom layer). For the input objects, select all the objects in the geometry (uncheck keep interior boundaries).

top structure

Duplicate Block 1 and change the z-position to $h_l + r_d + sp$

Duplicate Sphere 1 and change the position to $(w_lx/2, 0, h_l + r_d + sp)$

Duplicate Block 7 and change z-position to $r_d + sp$

Right-click Geometry 1 and select Booleans and Partitions|Intersection. For the input objects, select the sphere (sph6) and the last block (blk8).

Repeat the previous 3 steps changing each time the position of the sphere center with the following coordinates: $(0, w_{ly}/2, h_{l+r_d+sp})$; $(w_{lx}, w_{ly}/2, h_{l+r_d+sp})$; $(w_{lx}/2, w_{ly}, h_{l+r_d+sp})$.

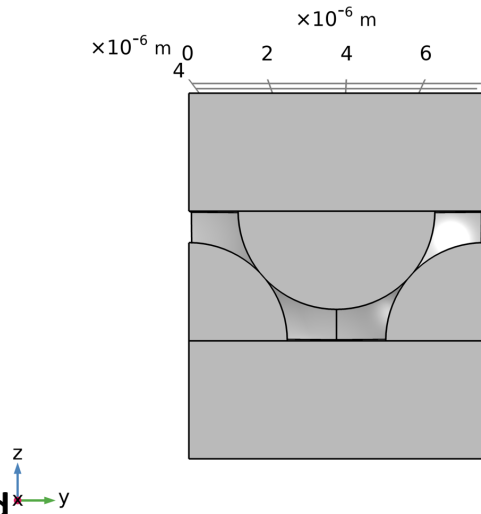
Right-click Geometry 1 and select Booleans and Partitions|Union (name: top layer). For the input objects, select all the objects of the top structure (uncheck keep interior boundaries)

Form Assembly

Under the Geometry 1 node select Form Union. Under Action select “Form an assembly”.

Under Pair type select “Contact Pair”. Click Build All.

In COMSOL, an **assembly** serves as a modeling feature that enables the **combination of various geometric parts** without merging them into a unified domain. This functionality is particularly beneficial for simulating **multiphysics interactions among components** that possess different materials, meshes, or physics interfaces, while **maintaining the independence of their boundaries**. The selection of **contact pairs** within an assembly establishes the **interactions between non-connected parts**, as their boundaries remain distinct.



Contact Pair

Under the Definitions sub-node select Contact Pair 1. Check “Manual control of selections”.

In the Source Boundaries selection, select the surface of the domes of the bottom layer.

In the Destination Boundaries selection, select the surface of the domes of the top layer.

Now the Contact Pair should appear as in figure.

Note: to see the geometry with this view click the Wireframe Rendering button in the Graphics window; then click the Go to Default View button

Materials

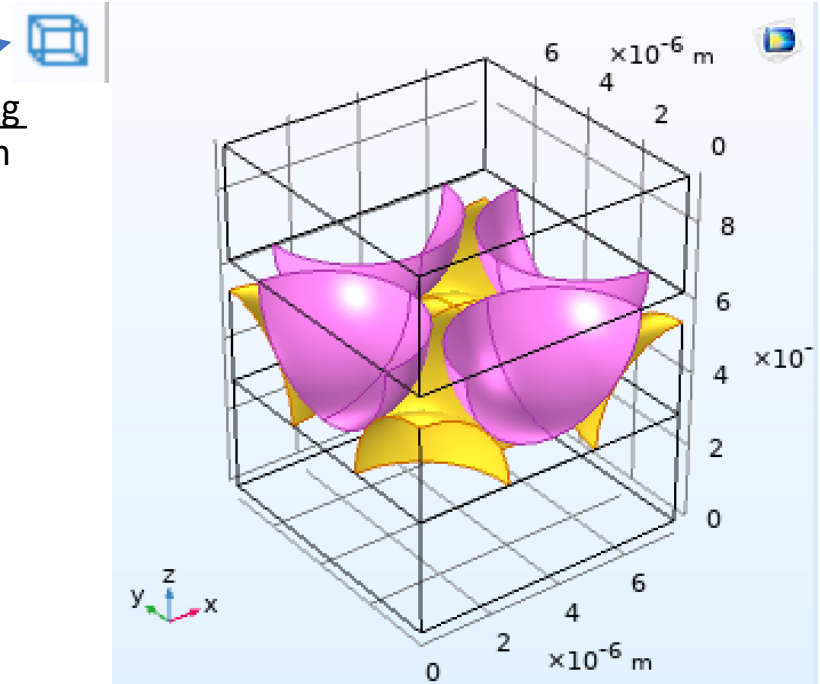
Right-click the Materials sub-node. Select “Add material from library”.

In the Add Material window double-click MEMS|Polymers|PDMS

Under the PDMS sub-node select “Young’s modulus and Poisson’s ratio”. Modify the values:

Young’s modulus to 2.6 [MPa]

Conductivity 0.5 [S/m]



Material: PDMS

- interlockedDomes.mph (root)
 - Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - Component 1 (comp1)
 - Definitions
 - Geometry 1
 - Materials
 - PDMS - Polydimethylsiloxane (mat1)
 - Basic (def)
 - Young's modulus and Poisson's ratio (Enu)
 - Solid Mechanics (solid)
 - Electric Currents (ec)
 - Multiphysics
 - Mesh 1
 - Solid mechanics
 - Step 1: Stationary
 - Solver Configurations
 - Electric currents

Output Properties

Property	Variable	Expression	Unit
Coefficient of thermal expansion	alpha_iso	9e-4[1/K]	1/K
Heat capacity at const Cp		1460[J/(kg*K)]	J/(kg*K)
Relative permittivity	epsilon_iso	2.75	1
Density	rho	970[kg/m^3]	kg/m^3
Thermal conductivity	k_iso ; kii	= 0.16[W/(m*K)]	W/(m*K)
Electric conductivity	sigma_iso	0.5	S/m

Expression:

- Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
- Component 1 (comp1)
 - Definitions
 - Geometry 1
 - Materials
 - PDMS - Polydimethylsiloxane (mat1)
 - Basic (def)
 - Young's modulus and Poisson's ratio (Enu)
 - Solid Mechanics (solid)
 - Electric Currents (ec)
 - Multiphysics
 - Mesh 1
- Solid mechanics
 - Step 1: Stationary

Output Properties

Property	Variable	Expression	Unit
Young's modulus	E	2.6[MPa]	Pa
Poisson's ratio	nu	0.49	1

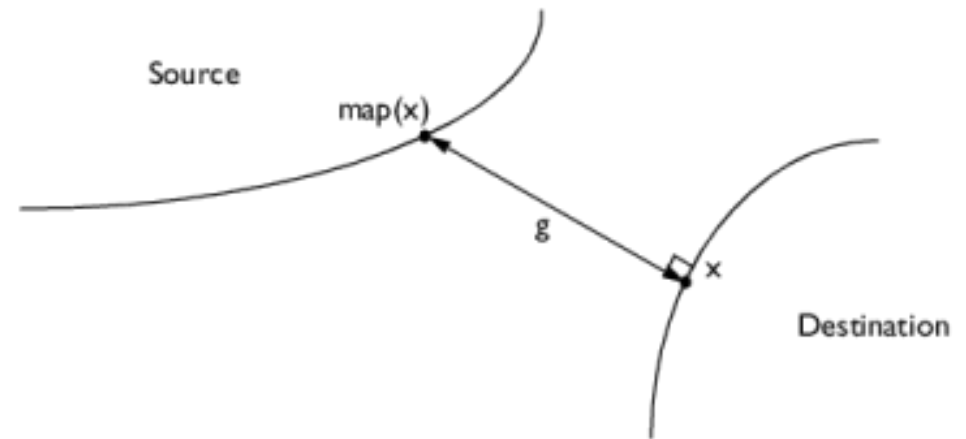
Expression:

Contact Modeling

Structural contact modelling is a **highly non-linear problem** { Source
Destination

Destination boundaries are constrained from penetrating source boundaries. Forces are computed over the destination boundaries

Destination boundaries must be meshed finer than source boundaries



Contact Modeling

- **Penalty method**

The penalty method for contact modelling in simulations is a numerical approach used to enforce contact constraints between interacting bodies. This method introduces a **stiffness parameter** (the **penalty parameter**) that **applies a force opposing the penetration** of one body into another. The greater the penetration depth, the larger the force exerted to correct it, mimicking the behaviour of contact without allowing unrealistic interpenetration. This approach is widely used due to its simplicity and effectiveness in handling contact problems, though it requires careful tuning of the penalty parameter to balance between accuracy and numerical stability.

$$T_n = \begin{cases} -p_n d_g & \text{if } d_g \leq 0 \\ 0 & \text{if } d_g > 0 \end{cases}$$

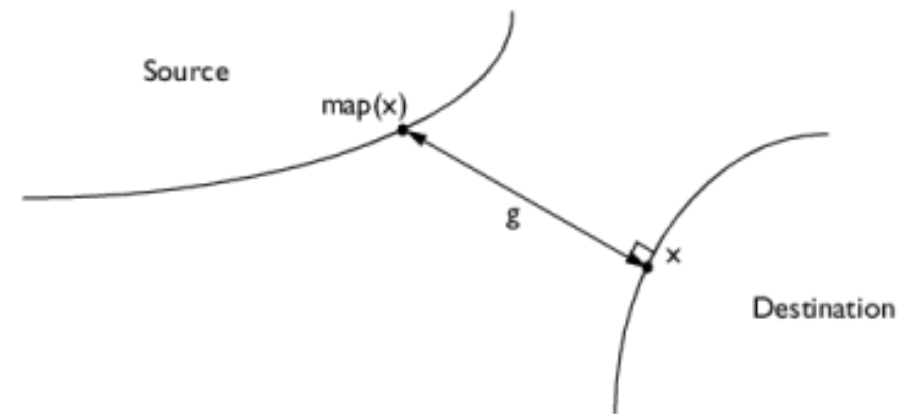
p_n : *penalty factor* d_g : *contact gap*
 T_n : *contact pressure*

Contact Modeling

Structural contact modelling is a **highly non-linear problem**: { Source
Destination

Selecting **source boundaries**:

- If there is a significant difference in the stiffness in the normal direction between the source and destination boundaries, select the **stiffer side as source**.
- When the contacting parts have approximately the same stiffness, consider the geometry of the boundaries instead. Make a **concave boundary the source** and a convex boundary the destination rather than the opposite.
- If one of the boundaries belongs to a part that is **rigid** it should be selected as the source boundary.

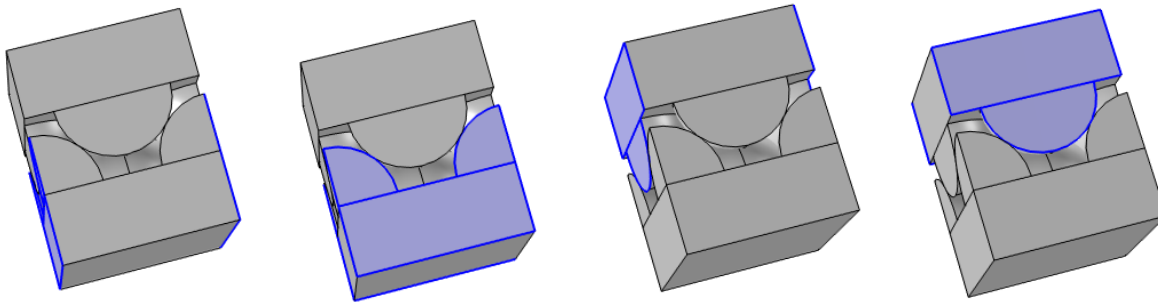


Solid Mechanics

Right-click Solid Mechanics sub-node and select the boundary Fixed Constraint for the bottom boundary.

Right-click Solid Mechanics sub-node and select Connection|Periodic Condition four times. Be sure that in Type of periodicity “Continuity” is selected.

In each periodic condition you have to select the external boundaries of opposite boundaries of the geometry for both top and bottom layers. See picture.



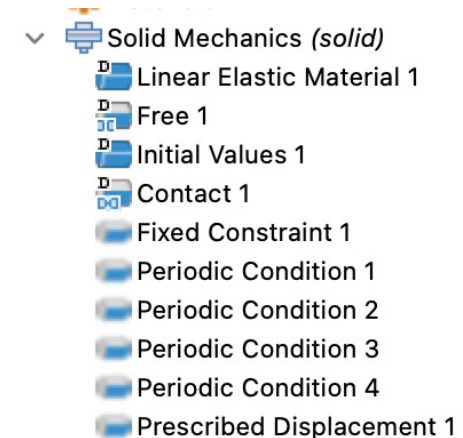
In Contact 1, verify that the contact Pressure Method select “Penalty”

Right-click Solid Mechanics sub-node and select the boundary Prescribed Displacement for the top boundary. Check all the boxes as prescribed and in the z-direction type “-disp”

The **continuity periodic condition** is used to model repeating structures efficiently by simulating only a representative portion of the domain.

It ensures that physical fields, such as displacement, temperature, or electric potential, are continuous across opposite boundaries.

This reduces computational cost while preserving the accuracy of the solution in systems with inherent periodicity.



Electric currents

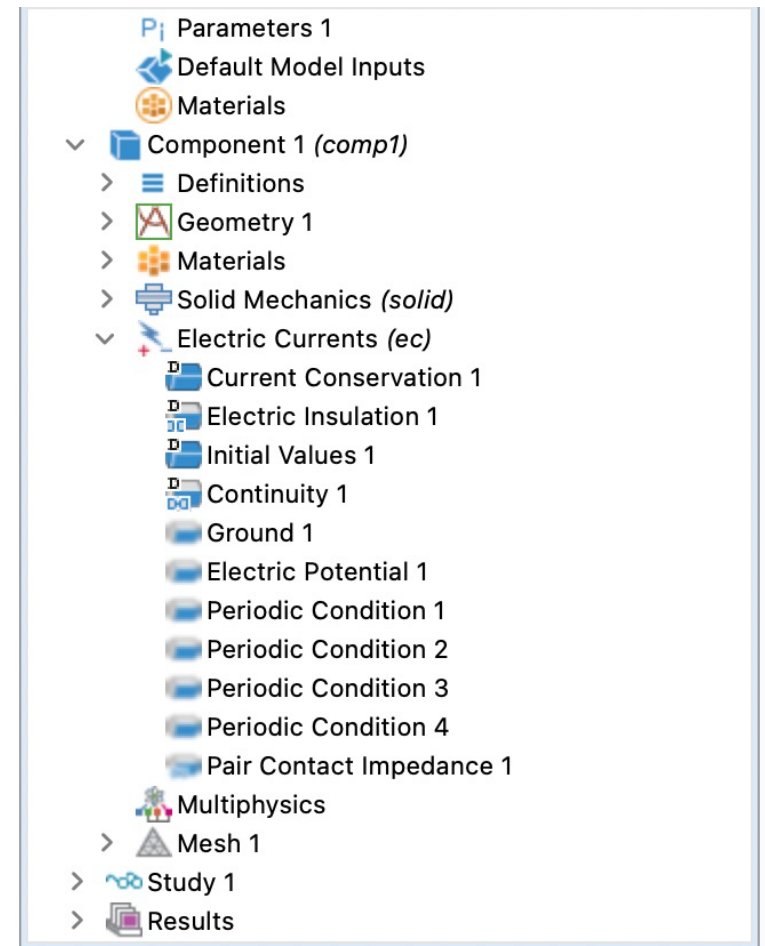
Right-click on the Electric currents sub-node and select Ground boundary condition. Select the bottom boundary.

Right-click Electric currents sub-node and select Electric potential boundary condition. Select the top boundary. In the Electric potential field type 1.

Right-click Electric currents sub-node and select Periodic Condition for four times. Perform the same selections did in solid mechanics.

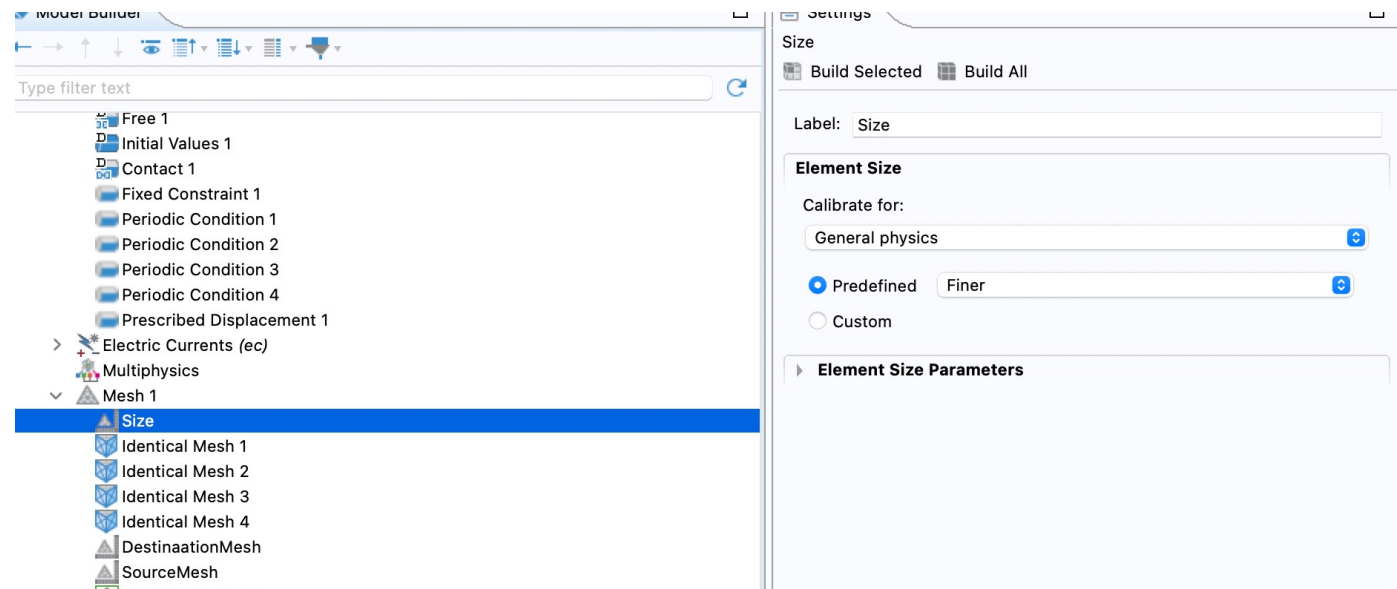
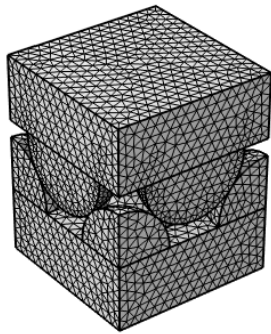
Right-click Electric currents sub-node and select Boundary Pairs|Pair Contact Impedance. In the Pair selection select Contac Pair 1.

Locate the Contact Impedance section. In the layer specification select Surface Impedance. Set the surface resistance to a very low value (e.g., $1e-15$). This will simulate an ideal electric contact.



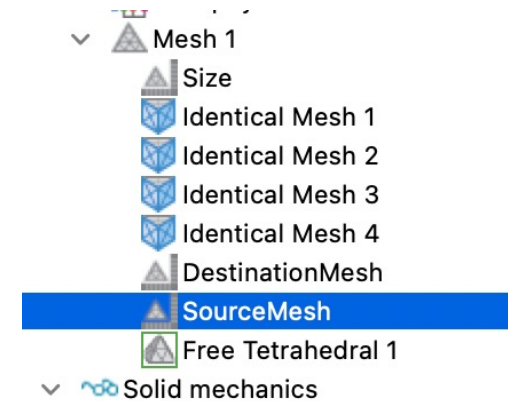
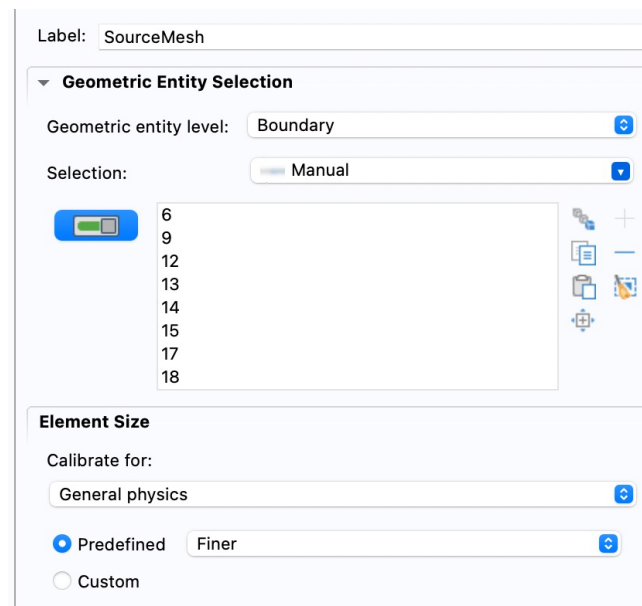
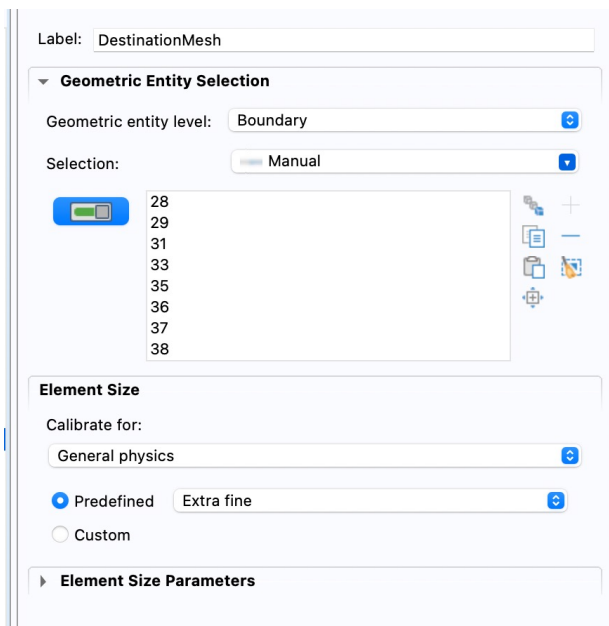
Mesh

Select the mesh sub-node and click Build All. You should see the predefined Mesh
Under “element size” check “predefined” and select “finer”.



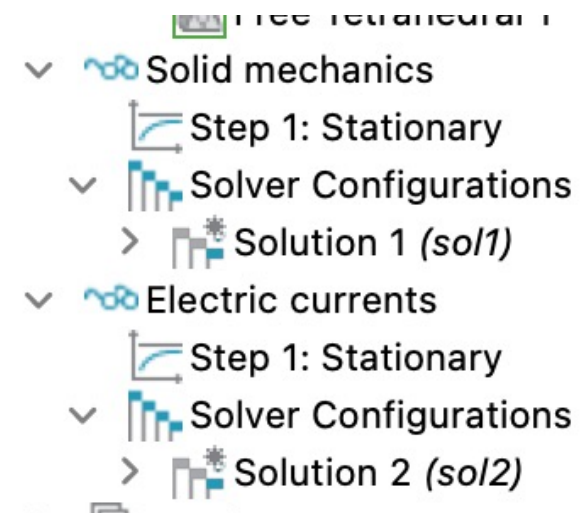
Mesh

Right-click two times and add “Size” with names “DestinationMesh” and “SourceMesh”. In both, select the geometric entity level at “boundary” and select The destination and source boundaries, respectively. In destination mesh use the element size as “extra fine” while in source as “finer”.



Studies

- Study 1-> solid mechanics
- Study 2 -> electric currents
 - Add as a second stationary study
- In both, include an auxiliary sweep in the displacement
- First run solid mechanics, then electric currents



▼ **Study Extensions**

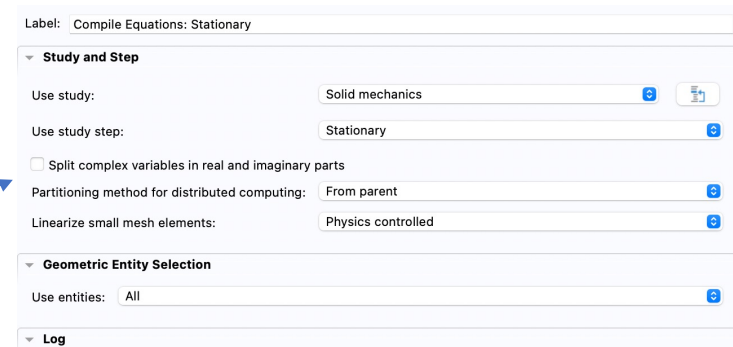
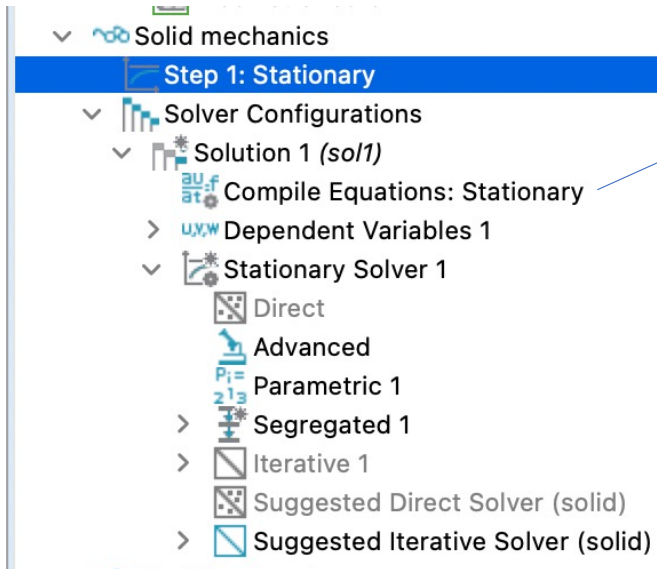
☒ Auxiliary sweep

Sweep type: Specified combinations

Parameter name	Parameter value list	Parameter unit
disp (initial displacement) ▼	range(displ_max/30, displ_max/30,displ_max)	m

Run continuation for: Last parameter

Study 1: Solid mechanics



Study 2: Electric currents

- Electric currents
 - Step 1: Stationary**
 - Solver Configurations
 - Solution 2 (sol2)
 - Compile Equations: Stationary
 - Dependent Variables 1
 - Displacement Field (comp1.u)
 - Electric Potential (comp1.V)
 - Stationary Solver 1
 - Direct
 - Advanced
 - Parametric 1
 - Fully Coupled 1
 - Iterative 1

Physics and Variables Selection

☐ Modify model configuration for study step

Node	Solve for	Equation form
Component 1 (comp1)		
Solid Mechanics (solid)		Automatic
Electric Currents (ec)		Automatic (Stationary)

Values of Dependent Variables

Initial values of variables solved for

Settings: Physics controlled

Values of variables not solved for

Settings: User controlled

Method: Solution

Study: Solid mechanics, Stationary

Solution: Current (Solution 1 (sol1))

Parameter value (disp (m)): Automatic (all solutions)

Compute electric solution on each step of the mechanical one.

Label: Compile Equations: Stationary

Study and Step

Use study: Electric currents

Use study step: Stationary

☐ Split complex variables in real and imaginary parts

Partitioning method for distributed computing: From parent

Linearize small mesh elements: Physics controlled

Geometric Entity Selection

Use entities: All

Log

Simulate the full sensor (not only the elementary unit)

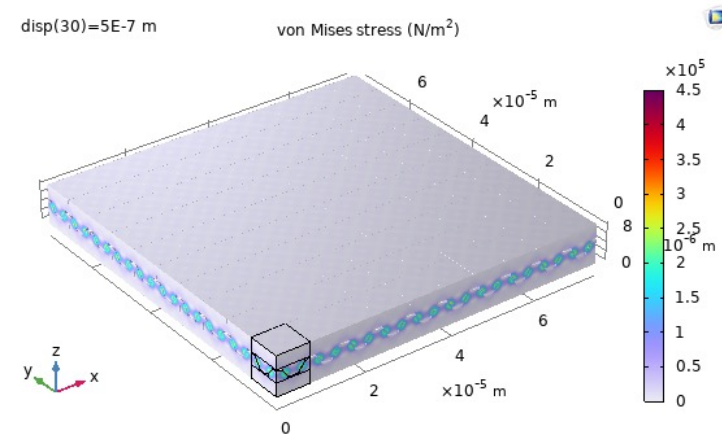
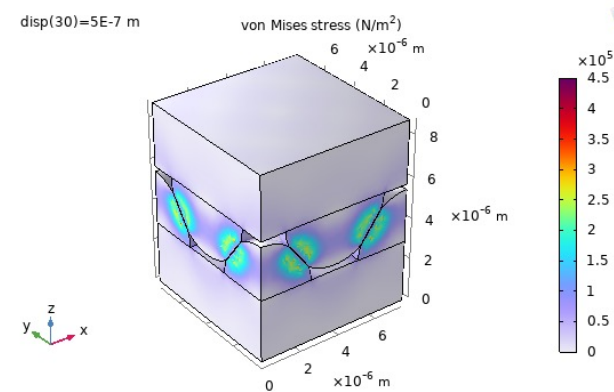
Array datasets

The screenshot displays the COMSOL Multiphysics software interface. On the left, the 'Model Builder' tree shows the hierarchy: 'Solution 2 (sol2)' > 'Compile Equations: Stationary' > 'Dependent Variables 1' > 'Displacement Field (comp1.u)' > 'Electric Potential (comp1.V)' > 'Stationary Solver 1' > 'Direct' > 'Advanced' > 'Parametric 1' > 'Fully Coupled 1'. The 'Results' section is expanded, showing 'Datasets' with 'Array 3D Solid Mechanics' selected. The 'Properties' panel on the right is configured for 'Array 3D Solid Mechanics' with the following settings: 'Label: Array 3D Solid Mechanics', 'Dataset: Solid mechanics/Solution 1 (sol1)', 'Input Method: Automatic', 'Array Size: Three-dimensional', 'x size: 10', 'y size: 10', 'z size: 1', 'Displacement Method: Automatic', and an 'Advanced' section. The 'Graphics' window on the right shows a 3D visualization of a red rectangular domain with a grid, representing the array. The axes are labeled x, y, and z, with dimensions in meters (m) and a scale factor of $\times 10^{-5}$ m. The bottom status bar shows 'disp (m)' and 'Contact area (m^2)'.

Results

Default von Mises stress on the Volume

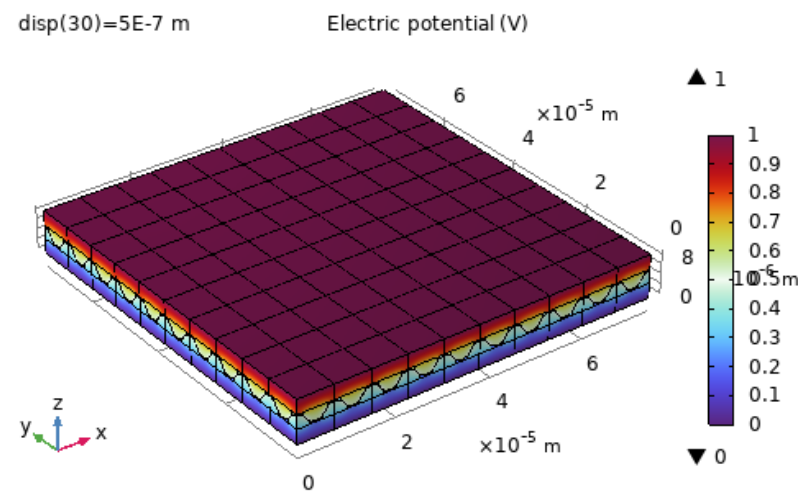
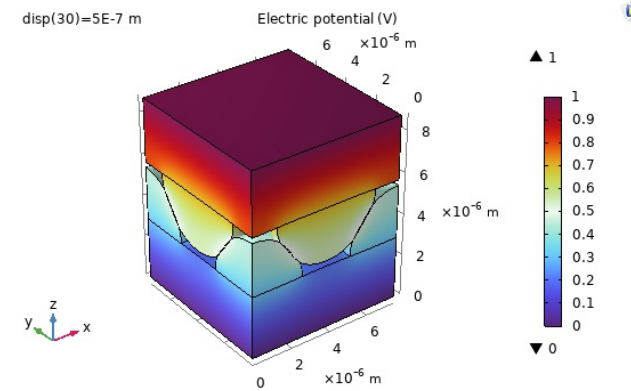
- Results
 - Datasets
 - Study 1/Solution 1 (sol1)
 - Array 3D 1
 - Views
 - Derived Values
 - Surface Integration 1
 - Surface Integration 2
 - Tables
 - Table 1
 - Table 2
 - Stress (solid)
 - Volume 1



Results

Electric potential on the volume

- Stress (solid)
 - Volume 1
- Electric Potential (ec)
 - Volume 1
- Electric Field (ec)

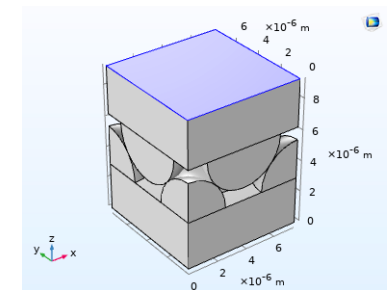


Results

Right click -> Derived values and add “Surface integration”, Select the top layer and add the expressions below to calculate pressure and current. Click on evaluate

The screenshot shows the COMSOL Multiphysics software interface. On the left, the 'Derived Values' menu is open, and 'Pressure-Current' is selected. On the right, the 'Surface Integration' settings are displayed. The 'Label' is 'Pressure-Current', the 'Dataset' is 'Electric currents/Solution 2 (sol2)', and the 'Parameter selection (disp)' is 'All'. The 'Selection' is set to 'Manual' with a selection of 26. The 'Expressions' table is shown below.

Expression	Unit	Description
ec.nJ	A	Current
-(solid.RFz)/(w_lx*w_ly)	N/m^2	Pressure

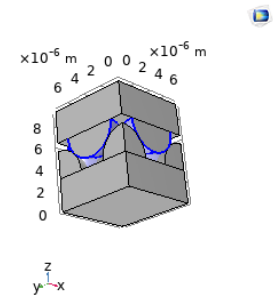


Results

Right click -> Derived values and add “Surface integration”, Select the boundaries of the destination (top layer) and add the expressions below to calculate the contact area. Click on evaluate

The screenshot displays the COMSOL Multiphysics interface. On the left, the 'Results' tree shows 'Derived Values' expanded, with 'Contact area' selected. The main window shows the 'Surface Integration' settings for 'Contact area'. The 'Dataset' is 'Solid mechanics/Solution 1 (sol1)' and 'Parameter selection (disp)' is 'All'. The 'Selection' is set to 'Manual' with a list of selected elements (28, 29, 31, 33, 35, 36, 37, 38). The 'Expressions' table at the bottom shows the expression 'solid.incontact' with unit 'm^2' and description 'Contact area'.

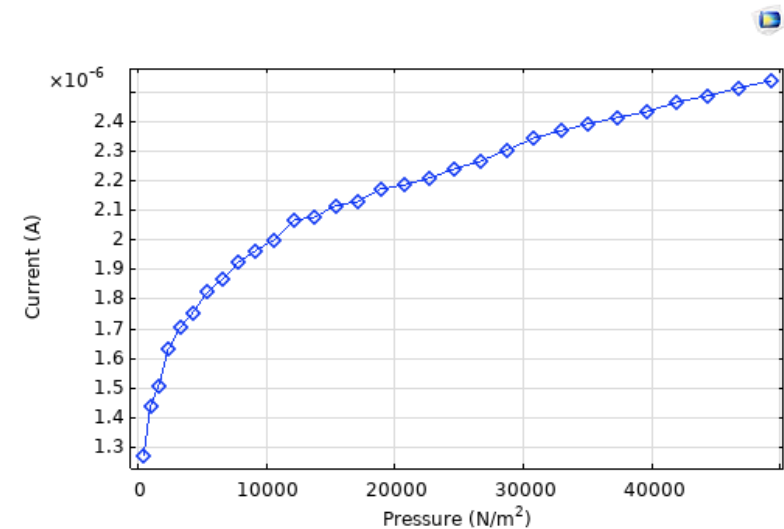
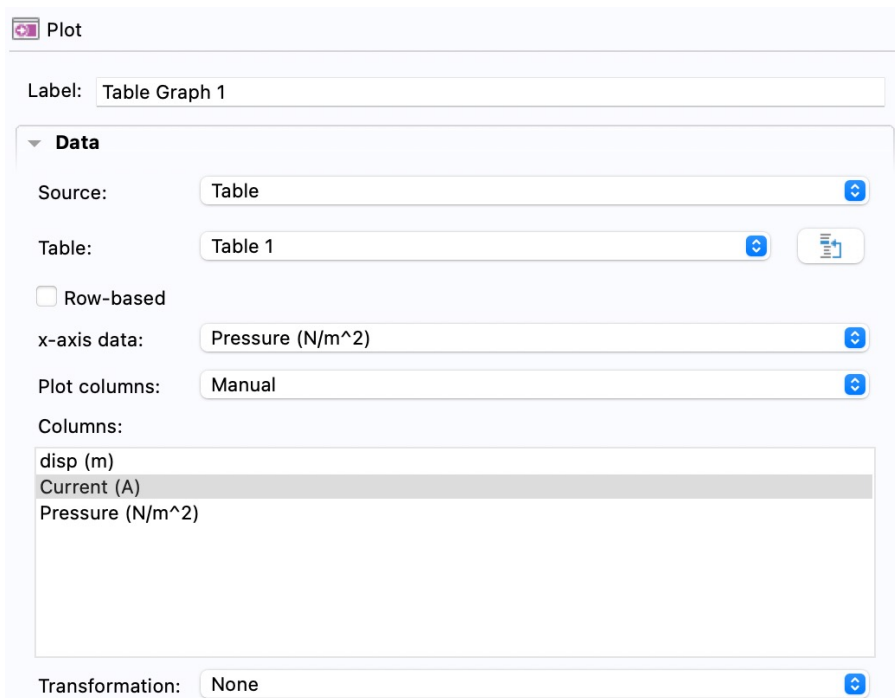
Expression	Unit	Description
solid.incontact	m ²	Contact area



Results

Under Results, right click -> 1D plot D group and add a Table Graph selecting Table 1 (select plot columns: manual)

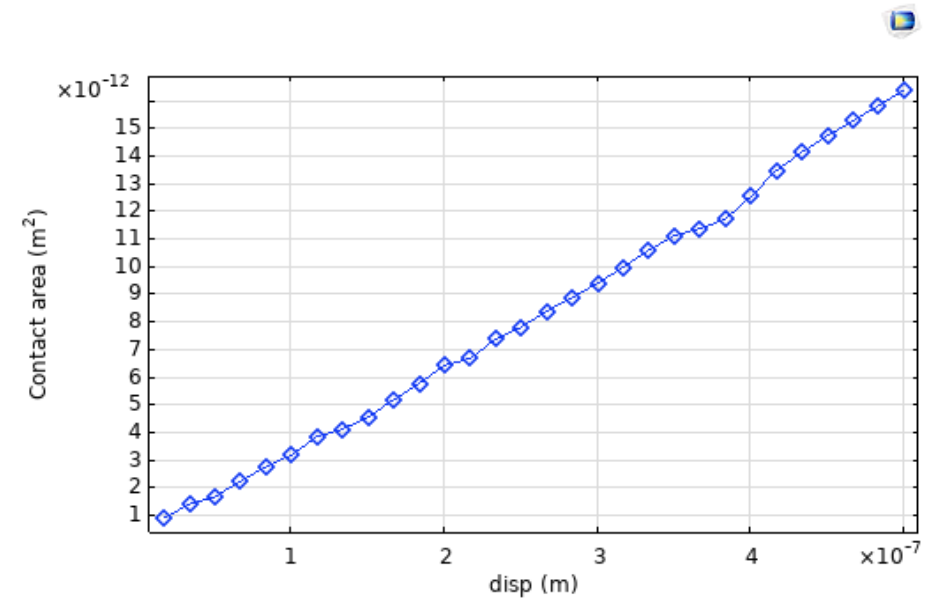
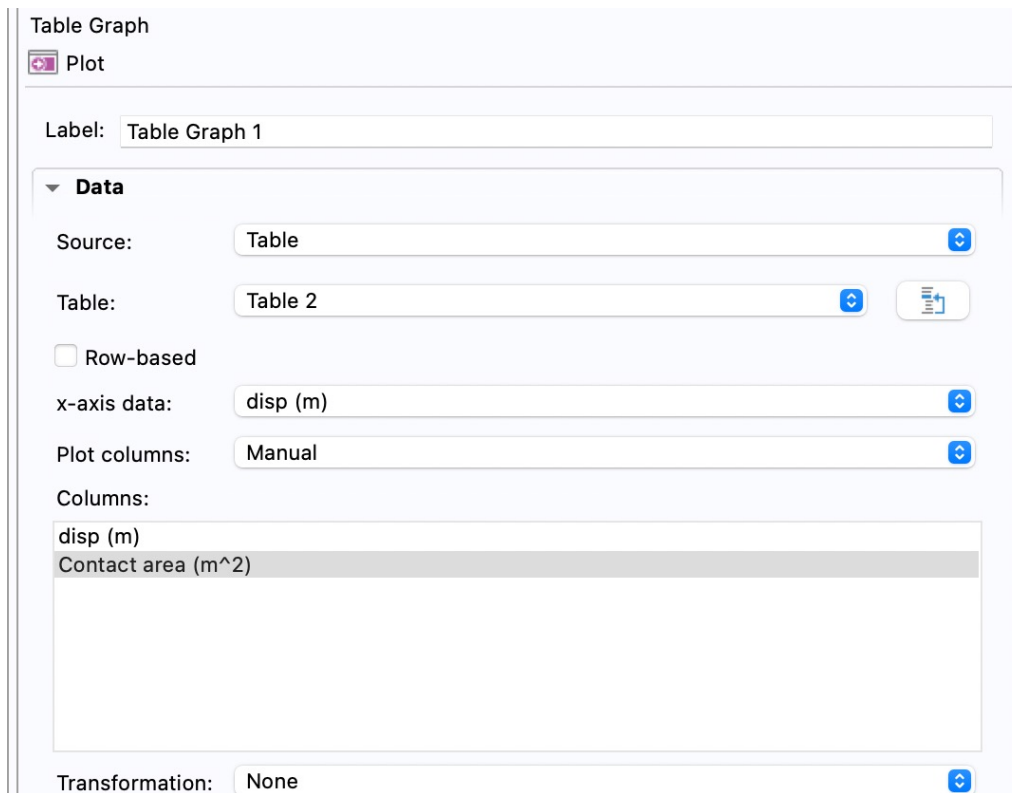
Plot Iout vs Pressure (see screenshot)



Results

Under Results, right click -> 1D plot D group and add a Table Graph selecting Table 2 (select plot columns: manual)

Plot contact area vs disp (see screenshot)



Results

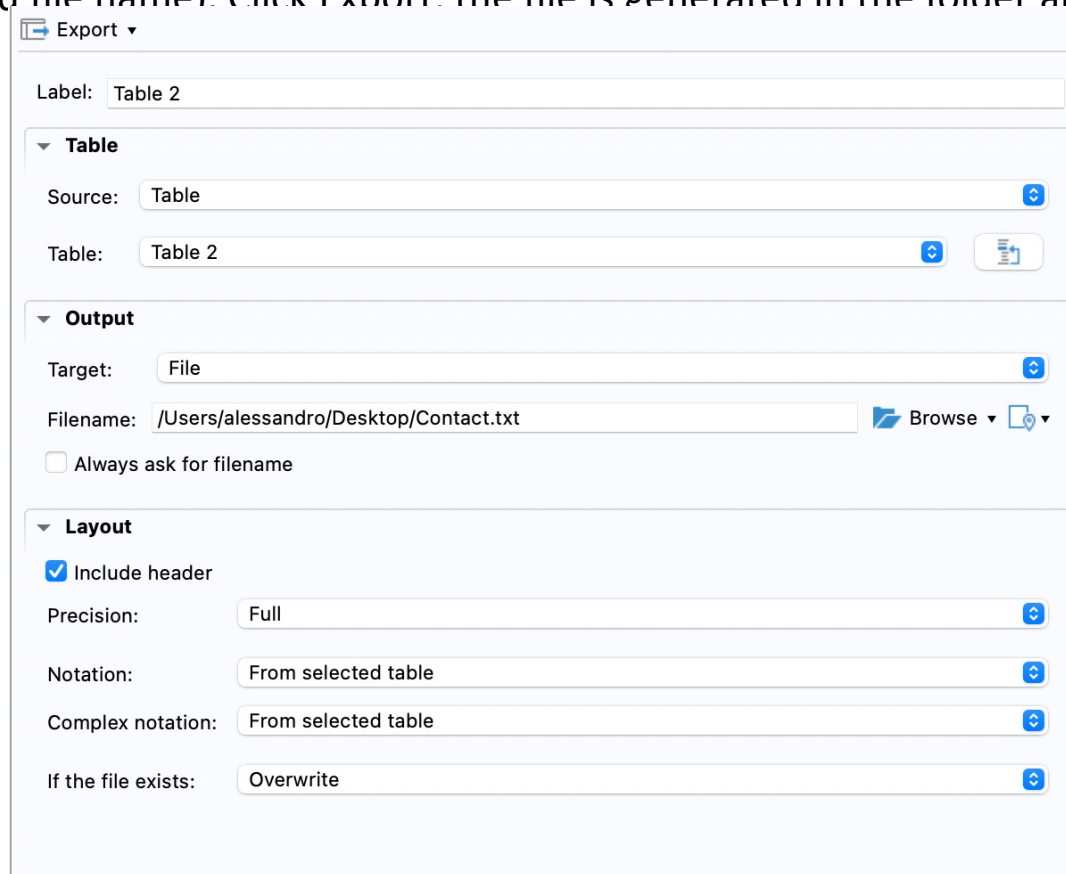
Under Exports, right click -> Table and select Table1. Add then a file name “PressCurr.txt” (click browse to add path and file name). Click Export, the file is generated in the folder and contains disp, pressure and lout.

The screenshot shows a 'Table' export dialog box with the following sections and settings:

- Export** (dropdown menu)
- Label:** Table 1
- Table** (section header)
 - Source:** Table
 - Table:** Table 1
- Output** (section header)
 - Target:** File
 - Filename:** /Users/alessandrognetti/Library/CloudStorage/OneDrive-UniversityofPisa/didattic (with a 'Browse' button)
 - ☐ Always ask for filename
- Layout** (section header)
 - ☒ Include header
 - Precision:** Full
 - Notation:** From selected table
 - Complex notation:** From selected table
 - If the file exists:** Overwrite

Results

Under Exports, right click -> Table and select Table2. Add then a file name “Contact.txt” (click browse to add path and file name). Click Export. the file is generated in the folder and contains disp and contact area.



Export ▾

Label: Table 2

▼ **Table**

Source: Table

Table: Table 2

▼ **Output**

Target: File

Filename: /Users/alessandro/Desktop/Contact.txt Browse ▾

☐ Always ask for filename

▼ **Layout**

☒ Include header

Precision: Full

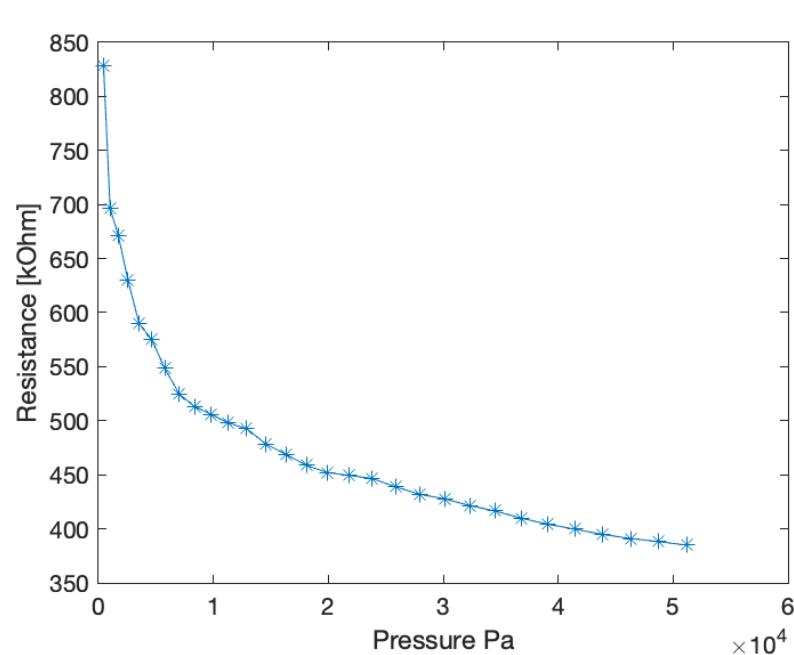
Notation: From selected table

Complex notation: From selected table

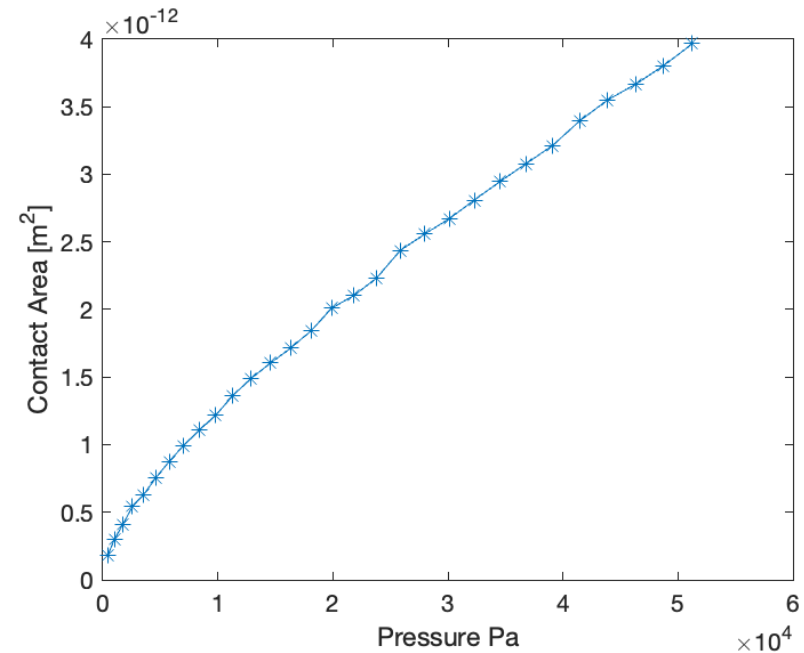
If the file exists: Overwrite

Results

Launch the Matlab file plot_data.mlx to plot the results of the simulation



Output of the sensor decreases non-linearly with pressure. Literature results confirmed!



Due to the increase in contact area
($R = \rho \cdot L / A$)

Final comment

What is the advantage of modelling an innovative sensors?

Model-based design: Test a design before expensive and work-demanding calibration experiments.

Example? If I want to to increase sensitivity?